AI Use Cases: Testing and Management

Exploring how artificial intelligence enhances both offensive and defensive security operations from automated vulnerability assessments and penetration testing to AI-driven incident management platforms and modern SIEMs.

5.1.4

SECAI+ PREP

Vulnerability Assessment

AI tools continuously find, analyze, and prioritize identified vulnerabilities by communicating with leading national and international vulnerability databases.

MITRE CVE

Common Vulnerabilities and Exposures — the global standard for publicly known cybersecurity vulnerabilities

NIST NVD

National Vulnerability Database — enriches CVE data with severity scores, impact ratings, and fix guidance

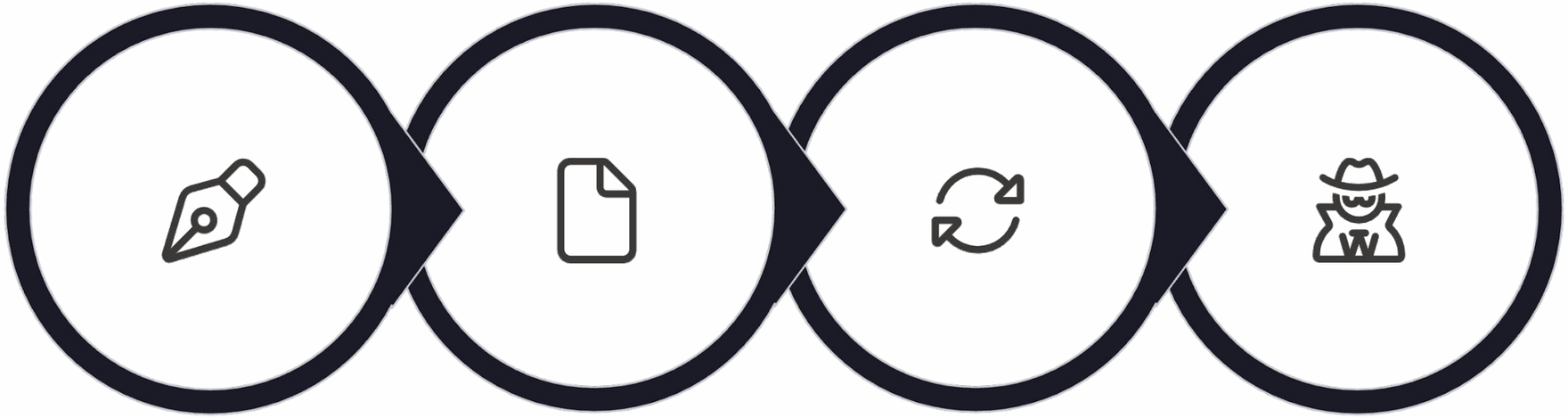
MITRE ATLAS

Adversarial Threat Landscape for AI/ML Systems tracks threats specifically targeting AI-powered environments

By integrating with these databases, AI facilitates proper identification of impact levels and exploitation likelihood, enabling security teams to prioritize remediation efforts effectively.

Automated Penetration Testing

AI-enabled penetration testing systems dynamically modify their attack strategies based on real-time responses from the target environment, simulating the adaptive behavior of real-world attackers.



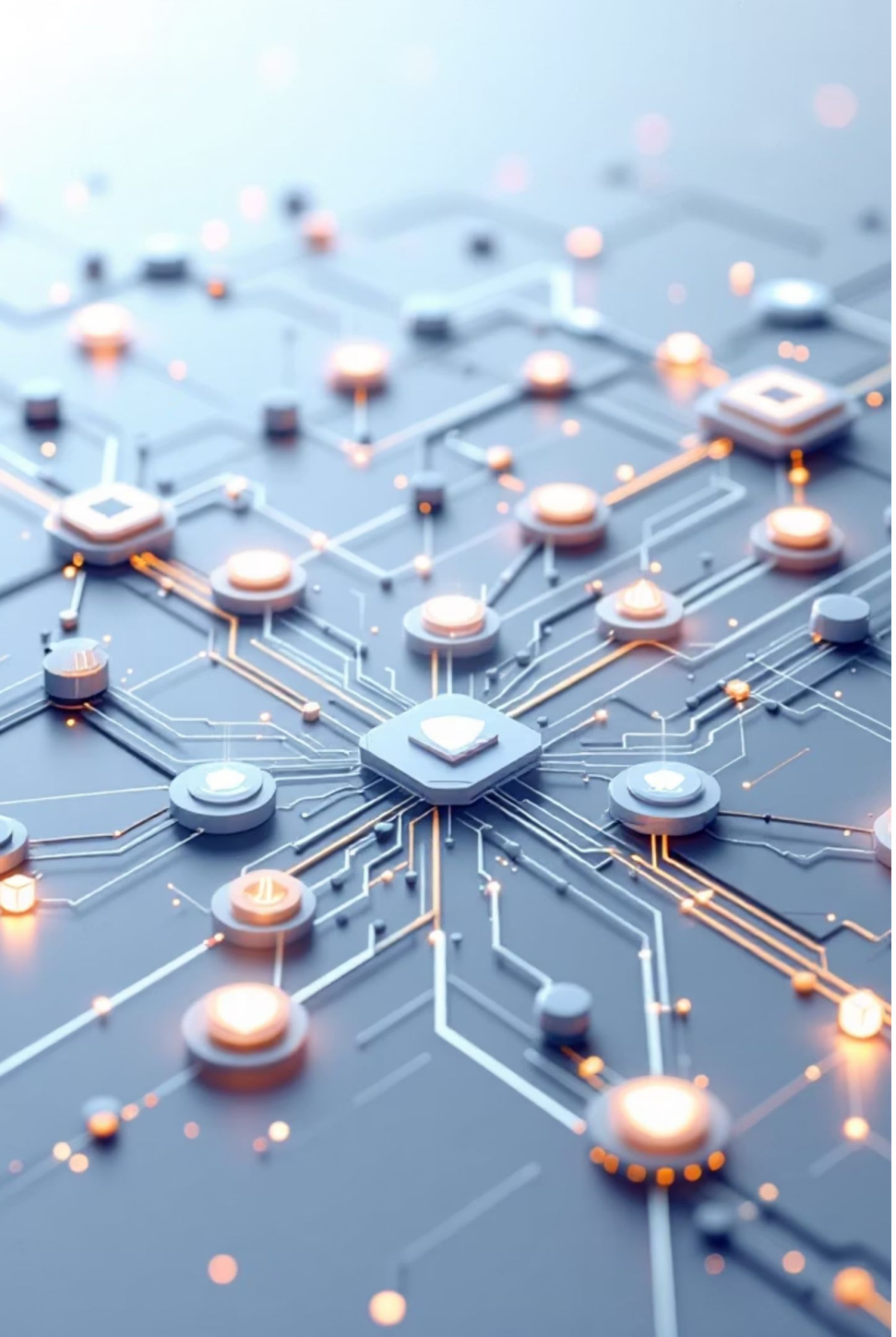
NMAP Scan

AI Analysis

Dynamic Adaptation

Realistic Simulation

This adaptive approach produces far more realistic and comprehensive security assessments than traditional static testing scripts, uncovering vulnerabilities that rule-based tools would miss.



AI-Powered Threat Modeling for Defense

How It Works

AI analyzes environment data, system configurations, and historical threat intelligence to generate tailored threat models and security improvement recommendations that are unique to each organization's risk profile.

→ Analyze Environment

Ingests configuration data, network topology, and asset inventory

→ Apply Frameworks

STRIDE and MITRE ATLAS enable rapid, structured threat model generation

→ Predict & Recommend

Forecasts future attacks and delivers organization-specific security improvements

Key Frameworks

STRIDE — Models threats across Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege

MITRE ATLAS — Specifically addresses adversarial threats targeting AI and machine learning systems in production environments

AI in Incident Management

AI transforms how security teams understand, track, and respond to incidents converting complex raw data into actionable intelligence at machine speed.

Human-Readable Summaries

Translates complex security event data into clear, concise summaries that enable faster incident comprehension and decision-making

Automated Tracking

Automates incident tracking and response timelines, reducing manual overhead and ensuring no critical step is missed during an active incident

AI-Authored Playbooks

Generates dynamic incident handling playbooks tailored to the specific threat type, environment, and organizational context

Auditability Required

All AI-generated outputs must maintain full transparency meaning that teams must be able to trace and verify how conclusions and recommendations were generated

Data Protection During AI Processing

The Challenge

When sensitive security data is passed to AI tools for analysis and reporting, there is a real risk of exposing confidential information including personally identifiable data, proprietary configurations, and classified incident details.

Controlled and sensitive data must be reviewed **both before and after** AI processing to ensure confidentiality is maintained throughout the entire workflow.

Required Safeguards

Mask Before Processing

Sensitive fields must be masked or tokenized before data is submitted to any AI tool

Redact Confidential Data

Personally identifiable and classified information must be redacted from datasets prior to AI analysis

Post-Processing Review

AI outputs must be reviewed to confirm no controlled data was inadvertently exposed or retained

Incident Management Platform Capabilities

Modern AI-powered incident management platforms go beyond simple alerting as they provide deep analytical capabilities that continuously improve an organization's security posture after every incident.

1

Timeline Analysis

Analyzes full incident timelines to reconstruct attack sequences and identify gaps in detection coverage

2

Response Evaluation

Compares actual response actions against expected response benchmarks to measure team effectiveness

3

Mitigation Review

Assesses whether existing mitigation controls performed as intended during the incident

4

Improvement Recommendations

Suggests configuration changes, targeted training programs, and new security controls based on incident findings

Key Takeaways

A summary of the core concepts from this lesson on AI use cases in security testing and management.

1. **AI Automates Vulnerability Assessment**
   AI tools continuously scan, analyze, and prioritize vulnerabilities by integrating with MITRE CVE, NIST NVD, and MITRE ATLAS databases

2. **Adaptive Penetration Testing**
   AI-enabled pen testing dynamically modifies attack strategies in real time, simulating realistic attacker behavior for more comprehensive assessments

3. **Threat Modeling with STRIDE & ATLAS**
   AI-powered threat modeling frameworks improve defensive posture by predicting environment-specific attacks and delivering tailored recommendations

4. **Incident Management Platforms**
   AI tracks, summarizes, and improves incident response, generating playbooks and post-incident recommendations automatically

5. **Modern SIEMs & New Risks**
   Sentinel + Security Copilot brings natural language to security investigations, while data poisoning and provenance risks must be carefully managed